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rather than observers, outcomes, or worlds). To illustrate the answers to these objections, a hypothetical Egalitarian toy model is described and defended that supports a quantitative simulation of the Aspect double delayed choice version of the EPR experiment. In the absence of a rigorous mathematical "no-go theorem," the dismissal of Egalitarianism is profoundly premature, and the *raison d'être* for decision theory is in doubt.

March 30, 2008

Studies in History and Philosophy of Science Part B:
Studies in History and Philosophy of Modern Physics

Dear editors,

I am pleased to submit my manuscript entitled “Egalitarianism offers a coherent alternative to decision theory as a solution to the problem of probability of the many worlds interpretation” for consideration of publication in Studies in History and Philosophy of Modern Physics. The subject matter of this manuscript is relevant to the core discussions of two international conferences in 2007, Many Worlds at 50 at the Perimeter Institute for Theoretical Physics in Ontario and Everett at 50 at the University of Oxford, and should be of interest to the journal’s readers, contributors and members of the Editorial Board, notably Prof. Bub, many of whom are experts in these topics.

I understand that submission of this manuscript implies that it has not been published previously, that it is not under consideration for publication elsewhere, and that if accepted it will not be published elsewhere in the same form, in English or in any other language, without the written consent of the publisher. I also understand that should the manuscript be accepted for publication, the Elsevier Ltd. ‘Transfer of Copyright’ agreement must be signed by me prior to publication.

I hope that the reviewing process finds the manuscript acceptable for publication in the journal.

Sincerely Yours,

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Egalitarianism offers a coherent alternative to decision theory as a solution to the problem of probability of the many worlds interpretation

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Abstract

Egalitarianism, loosely speaking, is the notion that probability is determined by “counting something” and setting probability in proportion to the count. This account of probability is central to classical statistical mechanics, yet has never been successfully implemented in quantum mechanics: hence the “problem of probability.” Advocates of the Deutsch-Wallace (DW) decision theoretic-based strategy to address this issue dismiss Egalitarianism on the grounds that it cannot be coherently defined. In this paper, the objections to Egalitarianism as put forth by Wallace (2007) and Greaves (2004) are reviewed and critiqued. The chief of these objections – that issues of continuity and infinity make counting incoherent – neglects various mathematical strategies that could be used to solve this problem. Indeed, Egalitarian concepts of probability have been explored recently in string theory and cosmology, undeterred by the technical difficulties of counting worlds in the landscape. The remainder of the objections rely on assumptions that are easily questioned or discarded (e.g., the assumption of counting Everettian branches rather than observers, outcomes, or worlds). To illustrate the answers to these objections, a hypothetical Egalitarian toy model is described and defended that supports a quantitative simulation of the Aspect double delayed choice version of the EPR experiment. In the absence of a rigorous mathematical “no-go theorem,” the dismissal of Egalitarianism is profoundly premature, and the *raison d’être* for decision theory is in doubt.

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1 Introduction

Loosely speaking, Egalitarianism is the idea that probability is determined by “counting something” and setting probability in proportion to the count. More specifically, it is the notion that the probability of X is equal to a ratio of the number of “X-like” states to the total number of “all possible” states. This idea formed the basis for the first sophisticated treatments of probability, which developed in about the seventeenth century through a mathematical

analysis of gambling (Hájek, Winter 2007). For example, consider the simple problem of determining the chance that a roll of a pair of dice will sum to 7. The number of “all possible” distinct die rolls is 36. If we count how many of these sum to 7, we find the answer is 6. Thus by Egalitarianism, the chance of rolling a 7 is $\frac{6}{36} = 16\frac{2}{3}\%$.

The notion of Egalitarianism remains central to the development of classical statistical mechanics, which is founded on the *fundamental postulate of classical statistical mechanics* (Huang, 1987):

An isolated system in equilibrium is found with equal probability in each of its accessible microstates.

This is also called *the postulate of equal a priori probability* because each microstate is treated equally: it is a “democracy of microstates,” so to speak. Consider, for example, the classical (non-quantum) treatment of thermodynamics, such as the classical kinetic theory of gases. Instead of dice, we are dealing with a collection of N distinguishable particles, the exact state of which is specified by the position p in three dimensional space and the momentum q (also a three dimensional variable) of each particle. A microstate therefore is a point in a $6N$ dimensional phase space, and it is microstates that we are counting. This concept of microstate is classical in the same sense that the pair of dice is a classical state: no weird quantum mechanical notions like superposition, nonlocality, etc. are necessary.

The derivation of classical thermodynamics from classical statistical mechanics requires the assumption, not only of the postulate of equal a priori probability, but also of an underlying mechanical theory of motion, i.e. Newton’s laws. Generally speaking, the aim of Egalitarianism is to provide a coherent and independent account of a *probability rule* P by derivation from a *fundamental mechanical theory* T supplemented by the postulate of equal a priori probability. For Egalitarianism to provide a *coherent* account, there must be a clear concept of microstate as a well defined, classical physical object that we have no trouble claiming as *real*.¹ For Egalitarianism to provide an *independent* account, T must be a classical mechanical theory from which P is derived – but without simply assuming the probability law itself. Once again, we may consider classical thermodynamics to provide examples of classical (non quantum) probability rules P .² Their derivations from classical statistical mechanics are based squarely on Newton’s laws as T (Huang, 1987), which tells us (for instance) not only how to define a microstate in a physically meaningful and relevant way, but also how to place microstates in groups that are physically relevant. For example, the “microcanonical ensemble” of microstates is defined

¹ what Bell (1988) might call a ‘beable’

² They are probabilistic in the sense that they are not necessarily followed; they are merely very very likely to be followed!

as the collection of microstates whose total energy must lie in the interval E and $E + \Delta E$ with energy defined classically. As another example, consider that the time evolution of a microstate – its trajectory in phase space – could not be calculated without Newton’s laws. The point here is that T is not simply a reworded statement of P ; they are so different, in fact, that one never stops to wonder whether a derivation of P from T might be circular. In general, the litmus test for the *coherency* of an Egalitarian account of a probability rule is whether we can answer the question: What are we counting? Likewise for *independency*: What is the fundamental classical mechanical theory T ?

Egalitarianism is generally considered to provide a coherent and independent account of probability in classical statistical mechanics, but not in quantum mechanics. This is related to – or it might even be said that it *is* – the problem of probability of quantum mechanics. Recently, there has been much interest in the application of decision theory, proposed by Deutsch (1999) and refined by Wallace (2007), to find a *non-Egalitarian* derivation of the Born rule as a solution to the problem of probability. One of the central tenets of Deutsch-Wallace (DW) decision theory is that Egalitarianism is incoherent and should be rejected. Unfortunately, unlike the derivation of the kinetic theory of gases from Newton’s laws, DW suffers unresolved charges of circularity.

It is the thesis of this paper that in quantum mechanics, a coherent and independent Egalitarian account of the quantum probability rule – the Born rule – can be established, and should be established, but has not *yet* been established. Some of the conceptual difficulties are already familiar to us from the study of classical statistical mechanics, the most notable being how to make sense of the notion of counting microstates when they are represented by points in a *continuous* phase space. In classical mechanics, the solution involves measure theory. In quantum mechanics, the solution is complicated by an additional conceptual difficulty: the only way to make a coherent Egalitarian account is for the microstates in question to represent *states of the observer*, not states of the thing observed.

The outline of the paper is as follows. In Sec. 2 there is a brief overview of the problem of probability in the Everett interpretation. Sec. 3 outlines a very basic Egalitarian interpretation of quantum probabilities. Sec. 4 points out that Egalitarianism is already being assumed by mainstream approaches to string theory. Sec. 5 outlines the best currently available non-Egalitarian account of quantum probabilities, Deutsch Wallace decision theory, and argues that its derivation of the Born rule is circular. Sec. 6 lists the objections to Egalitarianism by Wallace (2007) and Greaves (2004) that there is no coherent Egalitarian account of quantum probability, and provides rebuttals to each of these objections. Several additional objections are discussed and rebutted as well. Unfortunately, arguments for Egalitarianism are often met with the response: “prove it,” or, “I’ll believe it when I see it.” Therefore the answers to

the objections in Sec. 6 are used as a guide in Sec. 7 to build a hypothetical Egalitarian toy model. Sec. 8 defends the feasibility of this toy model to develop into a full blown independent Egalitarian account of quantum probability. We conclude that Egalitarianism remains a viable and coherent option that deserves further exploration, and whose success would obviate the need for DW decision theory.

2 The problem of probability

In quantum mechanics, if someone asks: “what exactly are we counting?” it turns out that no one has provided a satisfactory answer to this question. Probability is given by the Born rule, which states that the probability of the n^{th} out of N outcomes is in proportion to $|\phi_n|^2$ where ϕ_n is the amplitude associated with that outcome, as determined by the Schrödinger equation. The Born rule measure, however – as far as we know – does not represent the “count” of anything that could be called a microstate. The initial most obvious line of thought: “We are counting dots on a screen. What’s the problem?” quickly turns into the type of local hidden variable theory that is formally excluded by Bell’s theorem (Bell, 1988), and whose unworkability was expressed much earlier by Schrödinger (1935, page 488): “at no moment in time is there a collective distribution of classical states [microstates] which would be in agreement with the sum total of quantum mechanical predictions.” The search for an account of probability in quantum mechanics, Egalitarian or otherwise, is known as the problem of probability.

The problem of probability is felt most exquisitely when viewed through the relative state interpretation of Everett (1957), more commonly referred to as the many worlds interpretation (MWI). From the perspective of the MWI, a significant portion, perhaps the majority, of quantum “weirdness” is encapsulated in the problem of probability, which is currently regarded as the most difficult conceptual difficulty facing the MWI. Its resolution is therefore a major goal for MWI proponents.³ The underlying ontology of the MWI is that each Everettian branch (or world), in some sense, “happens;” each branch is in some sense equally “real.” One might expect, therefore, that we should be counting “Everettian worlds” or “Everettian branches” or some such thing, with each one being equally probable: “[I]t seems that all Everett can say is that each outcome occurs in exactly one branch, which, we might think, *will yield equal probabilities if any at all*” (Greaves, 2004, emphasis mine). Wallace (2007) states it similarly: “One of the most telling criticisms levelled at Everettians by their critics has always been their inability to explain why, when all outcomes objectively occur, we should regard one as more likely than the

³ This assumes, as most agree, that the preferred basis problem has been solved.

other.” This criticism refers to the fact that branches have equal probability only in the very special case that they are each assigned the same Born rule measure, and this is not generally the case. “The quantitative problem [problem of probability] is perhaps unproblematic in the case of equally weighted superpositions, but has been considered fatal in the case of unequally weighted superpositions” (Greaves (2004)). Simply counting Everettian branches gives the wrong answer, and if the Born rule measure is not the count of something, then it is unclear how or why it *is* a probability. As stated by Graham (1973), “it is extremely difficult to see what significance such a measure [the Born rule measure] can have when its implications are completely contradicted by a simple *count* of the worlds involved, *worlds that Everett’s own work assures us must all be on the same footing*” (emphasis mine).

3 The Egalitarian interpretation of quantum probabilities

The best way for the dedicated Egalitarian to make sense of the Everettian tree and its attendant problem of probability, is the following. Simply put, what is typically thought of as a *single result* – say, the n^{th} out of N possible results, realized with probability $|\phi_n|^2$ – should be equated with a *multitude of distinct microstates* $\bar{\phi}$, the number k of which is *proportional* to $|\phi_n|^2$.⁴ This is illustrated in Fig. 1. The basic hypothesis is that the “counting microstates” idea is the fundamental probability rule at the fine grained level (Fig. 1B), and that the Born rule probabilities are applicable at a more coarse grained level (Fig. 1A). We will refer to the basic idea outlined in Fig. 1 as the many microstates interpretation (MMI). The MMI is intended to be a rough sketch only; it encompasses models in which the Everettian tree in Fig. 1B is exact as well as models in which it is possible to increase the “grain” of Fig. 1B further.⁵

3.1 Litmus test # 1: what are we counting?

Relational quantum mechanics (RQM) (Rovelli, 1996) is an interpretation that is closely related to the MWI, one that focuses on the relation between the observer and the observed system. The key observation of RQM is the recognition that the state vector should not be conceptualized as yielding

⁴ For rational probabilities, finite k is sufficient; for real probabilities, infinite k is required, attained in the limit as the Everettian tree of Fig. 1B becomes infinitely fine grained.

⁵ The many microstates interpretation is similar to the many minds interpretation, e.g., of Albert and Loewer (1988).

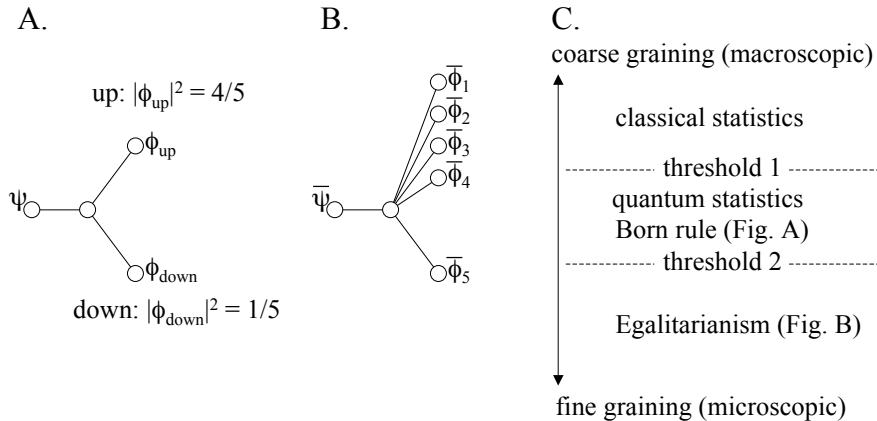


Fig. 1. The many microstates interpretation (MMI), illustrated by a spin-half measurement yielding the shown probabilities. A. Probability calculated by the Born rule. ϕ and ψ are wavefunctions in the usual sense. B. Probability calculated by outcome Egalitarianism. The bar indicates that the wavefunctions are replaced with classical states of the fundamental theory T (see text). C. Threshold 1 is the well-known quantum-to-classical transition that occurs at about the Planck scale. The MMI postulates a Threshold 2 at a finer scale, below which the Born rule is replaced by Egalitarianism.

(directly) the state of the observed system; rather it yields internal degrees of freedom *of the observer* that are correlated with the system under observation. This indicates to the Egalitarian that the answer to her question: what are we counting? is that *we are counting states of the observer, not counting states of the things observed*. The notion that we are counting Everettian observers will be referred to throughout this paper as *observer Egalitarianism*. On reflection, this point should be evident from a careful reading of Everett (1957) who notes that it is the *observer*, not the thing observed, that does the branching: “[W]ith each succeeding observation (or interaction), the observer state ‘branches’ into a number of different states.” It is the failure to recognize and appreciate this fact that the idea of counting dots on a screen – and thinking that each dot represents one state of the *observed* particle – does not work.

It should be emphasized that in RQM and in this paper, it is assumed that *any physical object* can play the role of the quantum mechanical “observer”

whether it is animate, inanimate, big, small, conscious or not. (Tegmark (2003) uses the term ‘self-aware state’, or SAS, to refer to a physical object that possesses consciousness.) All that is required is that there exist degrees of freedom internal to the observer that are correlated with degrees of freedom external to the observer. With the notion of observer Egalitarianism in mind, we are in a position to interpret an Everettian diagram as in Fig. 1 as follows. Each circle on the fine grained Everettian diagram Fig. 1B represents a physical object which can in principle be any physical object, to which we have appended the label of the ‘observer.’ Furthermore, the bar is meant to indicate that each circle is a *fully classical state*.

3.2 Litmus test # 2: What is T ?

In the above account of the MMI, we still have not specified what a microstate actually *is*, or what makes one microstate *distinct* from another. Furthermore, there is a certain circularity to its reasoning. The Born rule measure is justified as a probability because its value is proportional to the number of microstates associated with an experimental result; but conversely, the number of microstates associated with a result is defined as its Born rule measure. The underlying explanation for this circularity is that no T has been provided; the MMI provides a coherent, but not yet an independent, Egalitarian account of quantum probability. The ultimate hope of the Egalitarian is that it is possible to augment the MMI by a T in such a manner that it becomes an independent Egalitarian account of the quantum probability rule, i.e. the Born rule, in the same manner that Newton’s laws provide an independent Egalitarian account of the kinetic theory of gases.

If one accepts that the Everett interpretation tells us that all Everettian entities (worlds, observers, outcomes) truly exist – regardless of any associated quantum weights (probabilities) – then the Everett interpretation opens itself to the criticism that it is in fact totally devoid of content, because it tells us nothing more and nothing less than that *everything happens* (Greaves, 2007). In the context of observer Egalitarianism, this may be rephrased: “all possible observers exist.” Therefore, we will require of a fundamental law T that it prescribes how to construct a grand list of all possible pieces of matter, i.e. all possible observer-microstates $\bar{\psi}$. We might envision two forms that this list could take. The first we will label Ω , which contains a single copy of all possible $\bar{\psi}$ ’s. It should be clear from T how each element of Ω is *physically distinct* from any other element. The second we will label U , which can contain more than one copy of any given $\bar{\psi}$. If we imagine that U exists in the Platonic sense – “mathematical existence equals physical existence” – so it represents a “bird’s eye view” of Existence (what Tegmark (2003) might call a type 3 Multiverse), then we may propose a more detailed definition of Egalitarianism

of observers:

Definition 1 *Observer Egalitarianism: The relative probability associated with $\bar{\psi}$ is proportional to the number of copies of $\bar{\psi}$ in U .*

This will be compared in future sections to Egalitarianism of outcomes, branches, and worlds. We will define these to be distinct concepts, but will ultimately advocate a scheme in which Egalitarianism of observers, outcomes, worlds, and sub-branches (a variant of branch Egalitarianism) provide equivalent predictions. In this way, the ontologic merits of each individual type may be combined.

Another requirement of T will be that it allow us to establish and derive the correlations between internal degrees of freedom (recordings) of $\bar{\psi}$ to external degrees of freedom. For any observer we can query whether these internal recordings are consistent with the Born rule (or consistent with any proposition whatsoever, for that matter). If consistency with the Born rule could be shown in “most” elements of U (all but a set of measure zero), then the attempt to construct a “self-interpretation” of the MWI would likely be less problematic.

4 String theory: the Egalitarian landscape

Before discussing alternatives to Egalitarianism and the objections that it is incoherent, it should be noted that there are approaches to probability in string theory and cosmology that are inherently Egalitarian. Much of this work is done by cosmologists such as Alexander Vilenkin, Lisa Randall, Raphael Bousso, etc. in an attempt to characterize properties of the landscape of string theory. For example, Garriga and Vilenkin (2008) discuss the problem of making statistical predictions regarding constants of Nature, and define probability as follows: “When the observers in a given class perform an experiment, the class branches into subclasses who learn different information from the outcome of that experiment. The probabilities for the different outcomes are defined as the relative numbers of observers in each subclass.” Clearly, this notion is a statement of outcome Egalitarianism as illustrated in Fig. 1B. In another example, Karch and Randall (2005) use the notion of a d-brane – a hypersurface of dimension d embedded in a higher dimensional spacetime – to investigate why we observe 3 spatial dimensions, rather than the 10 or so dimensions of string theory. Assuming what they term the “relaxation principle” (RP), they “show that under conventional (but higher-dimensional) FRW evolution, a universe filled with equal *numbers* of branes and antibranes will naturally come to be dominated by 3-branes and 7-branes” (emphasis mine), where the RP is the assumption “that the branes with the biggest filling frac-

tion in the end-point of the universe’s cosmological evolution are the *most likely* branes to be relevant to the state in which we live” (emphasis mine). In other words, 3-branes (and 7-branes) are the most *likely* because they are the most *numerous* (or at least have the greatest measure). It would appear that the RP is based on the notion of counting branes: *brane Egalitarianism*.

It must be admitted that there are unsolved problems with the implementation of Egalitarianism in string theory. A prominent example discussed in Vilenkin (2007) is the measure problem: the problem of defining how exactly to count observers, branes, or other objects that exist within the landscape. The measure problem is essentially identical to Objection 1 below. A second example (which is not directly addressed in this paper) is the reference class problem, discussed in Garriga and Vilenkin (2008). However, there is no reason to believe that these problems cannot ultimately be overcome. The fact that Egalitarianism is considered a viable principle by some string theorists is, in and of itself, argument that it is premature to declare Egalitarianism to be incoherent.

5 The non-Egalitarian option: Deutsch Wallace decision theory

There have been a number of attempts to justify the Born rule that do not fit into the category of Egalitarianism. These are the “self-interpretation” arguments – sometimes claimed to elevate to the status of “derivation” – by, among others, Everett (1957); Hartle (1968); DeWitt (1970); Zurek (2005). These have been rejected as being circular, as discussed in Hemmo and Pitowsky (2007). The most recent of these is a proposal by Deutsch (1999), advocated by Wallace (2007), to “derive” the Born rule from the “non-probabilistic” axioms of quantum mechanics and decision theory. Deutsch-Wallace (DW) decision theory, like earlier attempts to derive the Born rule, suffers unresolved charges of circular reasoning. These arguments are well documented by many authors including Barnum et al. (2000); Price (2006); Baker (2007); Hemmo and Pitowsky (2007) and will not be recounted here. Several comments however will be made.

One of the essential assumptions of DW is that Egalitarianism does not solve the problem of probability, for if it did, there would be no need for DW in the first place. It is perhaps ironic that a central assumption of DW, *equivalence*, is given what appears to be an Egalitarian-like justification. Equivalence states: “if we know that two events have the same [quantum] weight [where quantum weight is defined as $|\psi|^2$], then we must regard them as equally likely” (Wallace, 2007). The proposed justification of equivalence, as noted by Price (2006), is that it possesses the “distinguishing symmetry of the Everett view, viz., that all possible outcomes of a quantum measurement are equally

real.” However, this is really a justification of Egalitarianism, not equivalence. Equivalence does not apply to *all* possible outcomes; rather, only to those possible outcomes that have equal Born rule measure. In other words, “all” of the outcomes are equally real – except for the ones that are not (which is usually the case!) – and the symmetry is lost. The distinguishing symmetry of Everett is Egalitarianism, not equivalence.

From the vantage point of the Egalitarian, an argument of circularity is formed around the observation that DW does not suggest anything that resembles an underlying fundamental classical mechanical theory T . On the other hand, it could be argued that the controversy of circularity surrounding DW arises because of a misunderstanding about what, exactly, DW is trying to achieve. In other words, the claims of its supporters – that it *derives* the Born rule from considerations of “pure rationality” (Wallace, 2007) – appear overstated, and seem to beg the question of what it means to derive something. Advocates claim that DW derives B from A, where B is the Born rule, and A are considerations of pure rationality. Critics charge circularity on the grounds that the Born rule is already assumed by A. How might we resolve this debate? One method might be to imagine a rational, intelligent, and reasonable agent who is fully ignorant of B, in fact has never even heard of B, and ask the question: if I were to hand the agent a list of assumptions A, sufficient to derive B, then what would be on that list? In this case we could imagine a nineteenth century physicist who knows classical mechanics but has never encountered quantum mechanics in any form, including the Born rule. Does the DW proof of the Born rule require the list A to contain this, or an equivalent, statement: *the frequencies of all prior observations match the Born rule predictions pretty darn well*? If the answer is yes, it would seem more fair to say that the goal of DW is not to derive the Born rule, but rather to prove that a rational agent must assign future probabilities according to a rule that agrees pretty darn well with frequencies of her past observations, *whatever they may be*. This is not a derivation of the Born rule because its arguments are actually independent of the Born rule. If the answer is no, then the claims of DW are truly remarkable: if only decision theory had been conceived earlier – say in 1850 – then the Born rule could have been discovered without anyone going to the trouble of actually doing any experiments. Of course, this is ridiculous, and it is not (by my understanding) what the advocates of DW are *actually* trying to say, even if it sounds like it sometimes.

One may well wonder why it should be so difficult to decide whether the Born rule is or is not assumed by DW. The answer may be that DW attempts to do the impossible: to assume the Born rule and not assume it at the same time. Consider that the “non-probabilistic” axioms, as stated by Saunders (2004), are the “assumptions of quantum mechanics that concern only the deterministic development of the state.” Among these assumptions are that the state of an object is represented by the symbol ψ , that ψ evolves unitarily,

that its evolution is governed by the Schrödinger equation (SE), and that the SE is confirmed by observation. Presumably we are supposed to assume all of this but *not* assume the probabilistic axioms, i.e. the *interpretation* of $|\psi|^2$ as a probability (the Born rule). Instead, we assume that there is this thing called the quantum weight which we define to be equal to $|\psi|^2$, and we assume equivalence: that if two outcomes have the same quantum weight, then they are equally probable. The problem with this is that quantum mechanics is nothing more and nothing less than a theory of probability, and it is not clear to me that it is possible or meaningful to separate the probabilistic from the non-probabilistic assumptions. To illustrate this point, imagine trying to do with classical mechanics what was done with quantum mechanics. (See Table 1.) Newton’s laws are nothing more and nothing less than a theory of the motion of matter; nevertheless, try to make sense of the directive to assume the “non-motion related” but not the “motion related” axioms of classical mechanics. The non motion related axioms include the assumption that the “state” of an object is described using the symbol x , and that the evolution in time $x(t)$ is governed by the equation $F = m\ddot{x}$. What we are *not* supposed to assume – because this is what we are trying to prove – are the motion related axioms, one example of which we will call the “momentum rule,” that the momentum $p = m\dot{x}$. We are careful that we do not simply define the momentum rule to be true; indeed, we do not even assume that x represents position. However we are supposed to assume something called equivalence: there is this thing called ‘weight,’ which is equal to $m\dot{x}$, and a rational agent would believe that if two objects have the same ‘weight’ then they have identical momenta. (We could envision similar derivations of the “position rule” position = x , the “acceleration rule” $a = \ddot{x}$, etc.) In this manner we claim to derive the motion related from the non motion related parts of Newton’s laws; in other words, we have *derived* the concept of momentum from considerations of pure rationality. Does this claim have merit? Of course not; at best, we can claim to have justified an interpretation of momentum. For the same reason, DW may succeed as a justification, but fails as a derivation of the Born rule.

6 Objections against Egalitarianism, and why they fail

We will evaluate the arguments against Egalitarianism presented by Greaves (2004, Section 5.3) and Wallace (2007, pages 327-329), as well as a few additional objections worthy of consideration. The process of responding to these objections will help the Egalitarian in her search for a T to render the MMI independent.

Assume: <i>equivalence</i>	(1)	$H\psi = i\hbar \frac{d}{dt}\psi$	$F = m\ddot{x}$
	(2)	the data confirm (1)	the data confirm (1)
	(3)	$\psi(t)$ = “state” at time t	$x(t)$ = “state” at time t
	(4)	$ \psi ^2$ = “weight”	$m\dot{x}$ = “weight”
	(5)	equal weight implies equal probability	equal weight implies equal momentum
without assuming:	(6)	any further definition of ψ	any further definition of x
Prove:	(7)	$ \psi ^2$ = probability (the Born rule)	$m\dot{x}$ = momentum (the “momentum rule”)

Table 1

Derivation of the “momentum rule” using reasoning analogous to the decision theoretic derivation of the Born rule. The derivations are circular because we cannot claim (2) without defining the symbols used in (1), i.e. without assuming (7).

6.1 The continuous branching objection

Objection 1 *Egalitarianism cannot be coherently defined because the number of branches is infinite.*

Greaves (2004) argues that “[B]ranching will in general be *continuous* – for instance, many of the interactions that lead to branching involve monitoring a system’s *position*, and position is a continuous variable. We cannot, therefore, *properly* talk of how many successors see spin-up, and Egalitarianism fails.”

Response to Obj. 1: *This problem was encountered and overcome in classical statistical mechanics through the application of measure theory. There is no reason the same methodology cannot be applied when counting Everettian branches (or observers, worlds, etc).*

Essentially the same objection has been encountered, and overcome, by classical statistical mechanics. As has already been discussed, the early accounts of probability dealt with discrete microstates as encountered in flips of the coin or in games of cards or dice. The microstates of classical statistical mechanics, on the other hand, are points in a continuous phase space, e.g. the $6N$ dimensional phase space Γ discussed in the introduction in which the exact position p and momentum q of N point-like particles defines one microstate. How can one define the *number* of states in the microcanonical ensemble – that is, the number of microstates with energy E ? There really *is no such number*; this is undisputed. But if we were to follow the logic of Greaves, then this would be grounds to reject Egalitarianism in classical statistical mechanics – that is, we would reject the fundamental postulate of classical statistical mechanics –

as incoherent. This is simply not justified. It is worth reviewing how this issue is addressed in classical statistical mechanics, because there is no reason that the same or similar methods cannot be used to address this issue in quantum Egalitarianism.

The main point to realize is that Egalitarianism does not require a “proper” or well-defined count of the number of microstates. What it does require for coherency is a proper definition of the *normalized* count of the number of microstates. Conceptually, the first step in this process is to define a *sample set* of the space Γ via a uniform distribution of some arbitrary finite number k of representative discrete points distributed uniformly throughout phase space. Probability is then calculated, via the equal a priori postulate, as the number of “representative” points falling within a given region divided by the total number of points. One simply requires that any given probability remains stable in the limit as the number of sampled points increases to infinity. In this way, the number of microstates is replaced with a measure over the space of microstates.

One of the difficulties encountered with the measure theoretic treatment of probability is that there is typically more than one definition of measure over phase space. This difficulty is illustrated by Bertrand’s paradox (Sklar, Winter 2007). Suppose that there is a factory that manufactures cubes, each with edge length between 0 and 1 unit, and each of random size. What is the probability that the edge length is in the interval $[0, \frac{1}{2}]$? Given a uniform distribution in the interval $[0, 1]$, the answer should be 50%. But consider next: what is the probability that the *area* of one cube face is in the interval $[0, \frac{1}{4}]$? To be consistent with our answer to the first question, the answer should be, again, 50%. However, if we assume an even distribution of points in the area interval $[0, 1]$, then we should expect the answer to be 25%, in contradiction to our first estimation. The answer lies in the recognition that the concept of an “even distribution of points in phase space” is incompletely defined until the measure is specified. So the question becomes: how do we know which measure is the “correct” measure? One general method is to select a measure such that the probability remains invariant under a transformation of interest. Ergodic theory, for example, can be argued to justify (in the right circumstances) the standard probability distribution measure by showing that it remains invariant under time evolution, as we would expect for a system in equilibrium (Sklar, Winter 2007).

Of course, one could always simply justify the choice of measure over which to apply the equal a priori postulate, as the one that results in the predictions that best fit the data, as described by Rushbrooke (1949): “For the present we must regard the validity of [thermodynamic laws] as ultimately resting on the agreement of theoretical results deduced from it with experimental data ... In the absence of any indication to the contrary, we can only assume that

‘all conceivable different micromolecular states of the assembly (complexions) corresponding to the same values of E, V , and N , are equally probable.’ This hypothesis underlies the whole of statistical mechanics and ... must, for the present, be regarded as justified a posteriori, by the success of the theory based on it.”

This challenge, that the equal probability postulate can really only be justified a posteriori and not a priori, was in fact considered by the early probability theorists. How does one know whether a coin is truly fair? Although Laplace argued for a justification a priori, via his “Principle of Indifference,” the truth is that a coin cannot be said to be fair until it has been tested – and so the justification is a posteriori after all. This remains an outstanding philosophical problem in the understanding of classical (non-quantum) probability, a solution to which is not sought in this paper.

Getting back to quantum mechanics, it is the hope of the Egalitarian to define a “phase space” for Everettian objects of some kind (branches, worlds, outcomes, observers ...) and then to address the issue of defining a measure over phase space using the same set of measure theoretic tools, and the same strategy of attack, that have been applied to classical statistical mechanics. It would be a remarkable achievement to do this: it would put the ontologic understanding of quantum probabilities on equal footing with that of classical probabilities. It is worth considering how it may be possible to solve the problem of the preceding paragraph and thereby end up on even better footing. Given a specific T from which to derive the Born rule, there may be some obvious or “natural” counting method that could possess some arguably a priori justification. This could be the case if points in Everettian phase space turn out to be discrete; or if they are continuous but countable; or if they are uncountable yet possess some otherwise most natural measure. One might envision that a philosophic justification for a measure defined a priori, for example, could arise from Tegmark’s concept of the Everettian multiverse that all possible mathematical structures exists; if a space over these structures could be defined, and a natural measure defined, then they would be a natural substrate on which to apply the equal a priori postulate.

Thermodynamics is not the only theory that would be (inappropriately!) abandoned by the logic of Greaves’ objection. Consider an example from quantum mechanics: the Feynman path integral (FPI) technique (Feynman and Hibbs, 1965), which requires us to add amplitudes of particle paths as if they were discrete, when in fact any individual path can be continuously deformed into any another. How is it conceivable to enumerate “all possible paths” when deformation of one path into another is a continuous process? The answer is that for the practical implementation of the FPI, we typically consider a finite number K of “representative” paths. Suppose we double the number K (by doubling our sample number, which we are free to do). This would have the

effect that the absolute value of the sum of the amplitudes over K paths approximately doubles. So the sum is not well defined! But what is well defined is the *normalized* amplitude. We are of course free to consider the case that K approaches infinity; and we require that the resulting probabilities are *stable in the limit as k approaches infinity*. Once again, the logic of Obj. 1 would have us believe that we cannot properly talk about the sum over paths, and the FPI fails. Clearly this is not the case; Obj. 1 is premature.

6.2 The ‘where to count?’ objection

Objection 2 *Egalitarianism cannot be coherently defined because there is no well-defined location to count Everettian branches.*

Wallace (2007) objects that Egalitarianism requires that the agent “will be faced with the impossible task of calculating how much branching will occur across the entire lifetime of the Universe ... to weigh up the value, now, to him of carrying out a certain act.” Branch counting cannot be defined because there is no “point after which the dust has settled and the value to an agent of consequences can actually be assessed.”

Response to Obj. 2: *This objection applies to branch Egalitarianism but not to the better-motivated observer Egalitarianism.*

The answer to Obj. 2 has already been well-articulated by Price (2006, footnote 2): “It is worth emphasising that what is excluded by this consideration is Egalitarianism *about branches*. A different view would be Egalitarianism *about outcomes* – i.e., the view that ... we should accord equal weight to all the outcomes to which the Born rule ascribes non-zero probability. The lack of even an approximate way to define the number of descendants is bad news for branch-Egalitarianism, but excellent news, at least *prima facie*, for outcome-Egalitarianism – for it suggests that among all the kinds of futures to which the Everett world commits us, there is no sense in which some occur “more often” than others. ... In this sense at least, then, Egalitarianism does seem to remain in play – an alternative decision policy whose exclusion needs to be justified”

Let us define explicitly these two types of counting. Figure 2A illustrates the difference between them.

Definition 2 *Branch Egalitarianism.* *The probability of a given outcome is proportional to the number of branches associated with that outcome at the end of the experiment.*

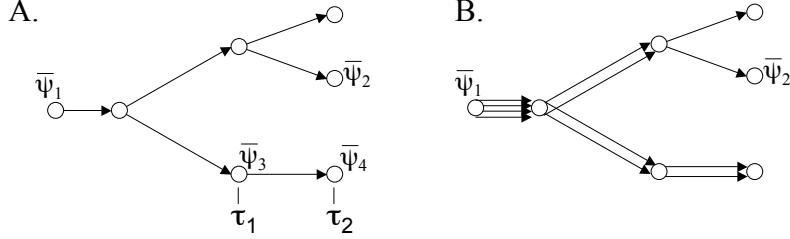


Fig. 2. A. The Everettian tree: counting outcomes versus counting branches. What is the probability that state $\bar{\psi}_1$ will evolve to state $\bar{\psi}_2$? By branch Egalitarianism, the answer is $\frac{1}{3}$ (there are, at the end, three branches); but by outcome Egalitarianism, it is $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. Note that outcome Egalitarianism (but not branch Egalitarianism) is transitive: each node is multiplied in succession. B. The Everettian sub-tree. Each branch in A is replaced by a number of sub-branches according to rules outlined in the text. The probability that state $\bar{\psi}_1$ will evolve to state $\bar{\psi}_2$ is $\frac{1}{4}$. In general, sub-branch and outcome Egalitarianism yield equivalent results.

Definition 3 *Outcome Egalitarianism.* Given a node that splits into N branches, each branch receives $1/N$ of the share of probability measure from its parent branch.⁶

From Figure 2A we see, first of all, that Wallace's objection is an apt criticism of branch Egalitarianism: the probability that state $\bar{\psi}_1$ will evolve to state $\bar{\psi}_3$ is $\frac{1}{2}$, since there are two branches at the slice defined by τ_1 . However, the probability that state $\bar{\psi}_1$ will evolve to state $\bar{\psi}_4$ is $\frac{1}{3}$, since there are three branches

⁶ In general terms, in outcome Egalitarianism, the probability of evolution from state $\bar{\psi}_i$ to $\bar{\psi}_j$ through a series of K states equals $\prod_{k=1}^K \frac{1}{N_k}$ where N_k is the number of branches that split from the k^{th} state. If we allow fusion as well as splitting in the Everettian tree, then it is possible for there to be more than one distinct path from $\bar{\psi}_i$ to $\bar{\psi}_j$. In this case, the total probability of evolution from $\bar{\psi}_i$ to $\bar{\psi}_j$ equals the sum of the probability associated with each individual path.

at the slice defined by τ_2 . But the probability of evolution to $\bar{\psi}_3$ ought to be the same as the probability of evolution to $\bar{\psi}_4$, since $\bar{\psi}_3$ evolves to $\bar{\psi}_4$ with certainty; hence we have a contradiction, and branch Egalitarianism, as described, is incoherent. This could be rectified by asserting that probabilities are to be calculated using a time-slice infinitely downstream. But this too is problematic; it would seem to require the assumption that all branching eventually terminates – perhaps suggesting some sort of entropic heat death of the universe – and we may not be inclined to commit to this assumption. In spite of Wallace’s objection, we cannot completely rule out that some modification may render branch Egalitarianism coherent. Nevertheless in this paper we will reject branch Egalitarianism in its simple form. We will subsequently resurrect it via a modification – sub-branch Egalitarianism – described shortly.

What about outcome Egalitarianism? The outcome Egalitarian can view the Everettian tree as representing a giant classical stochastic state machine. We might imagine placing each individual state $\bar{\psi} \in \Omega$ on a sheet of paper, with various arrows going from one state to another as in Figure 2A to indicate which states can evolve into which other state(s). Each state occurs in the diagram only once. Probabilities of evolution from one state to the next are assumed given by outcome Egalitarianism. If we begin the state machine in any arbitrary state, make N state transitions, and calculate the percentage $P(\bar{\psi})$ of iterations that the state machine occupies each individual state $\bar{\psi}$, then (in the limit as N approaches infinity) it is intuitive to interpret $P(\bar{\psi})$ as a probability measure. (If the number of states is infinite, then it may be necessary to impose certain additional assumptions, such as ergodicity and metric indecomposability, as Sklar (Winter 2007) discusses for the classical statistical mechanical case.)

Earlier (Sec. 3) we discussed U as being a collection of all possible observers and used it to define observer Egalitarianism. If we postulate that the following is true of U :

Axiom 1 *Begin the state machine in an arbitrary state $\bar{\psi}$, and make N state transitions. Probability of each transition is governed by outcome Egalitarianism. At each iteration, we add one copy of whichever state $\bar{\psi}$ is occupied by the state machine to an ensemble. In the limit as N approaches infinity, the ensemble constructed in this manner is U .*

then the number of times that a state $\bar{\psi}$ occurs in U is proportional to $P(\bar{\psi})$ (defined above). In this way we generate an equivalence between outcome Egalitarianism and observer Egalitarianism:

$$\begin{array}{c}
\text{observer Egalitarianism} \\
\Updownarrow \text{Axiom 1} \\
\text{outcome Egalitarianism}
\end{array} \tag{1}$$

Let us finally consider this modification of branch Egalitarianism:

Definition 4 *Sub-branch Egalitarianism.* *The probability of a given outcome is proportional to the number of sub-branches associated with that outcome at the end of the experiment.*

The concept of a sub-branch is illustrated in Figure 2. In Figure 2B, an Everettian “sub-tree” is constructed from the Everettian tree (Figure 2A) according to the following two rules. First, we allow there to be instances with more than one arrow (sub-branch) directly connecting any two $\bar{\psi}_i$ to $\bar{\psi}_j$. Second, we require that the total number of sub-branches flowing into a state equals the total number of sub-branches flowing out of the state. (In this way we have a sort of conservation law, equivalent to the notion that each inflowing sub-branch identifies with one outflowing sub-branch.) Interestingly, we see that sub-branch Egalitarianism can be made equivalent to outcome Egalitarianism if we make the following assumption:

Axiom 2 *Given any two states $\bar{\psi}_i$ and $\bar{\psi}_j$ immediately downstream of state $\bar{\psi}$, then the number of sub-branches from $\bar{\psi}$ to $\bar{\psi}_i$ equals the number of sub-branches from $\bar{\psi}$ to $\bar{\psi}_j$.*

That is, the sub-branches are divvied up evenly between the all of the states immediately downstream of it. This yields:

$$\begin{array}{c}
\text{outcome Egalitarianism} \\
\Updownarrow \text{Axiom 2} \\
\text{sub-branch Egalitarianism}
\end{array} \tag{2}$$

In summary, by answering Objection 2 we have been compelled to differentiate between different types of Egalitarianism, and in the process have identified two axioms that yield the very interesting result that *outcome, observer, and sub-branch Egalitarianism are mathematically equivalent.*

6.3 The Free Will objection

Objection 3 *Outcome Egalitarianism yields too much control over probabilities to the agent with free will.*

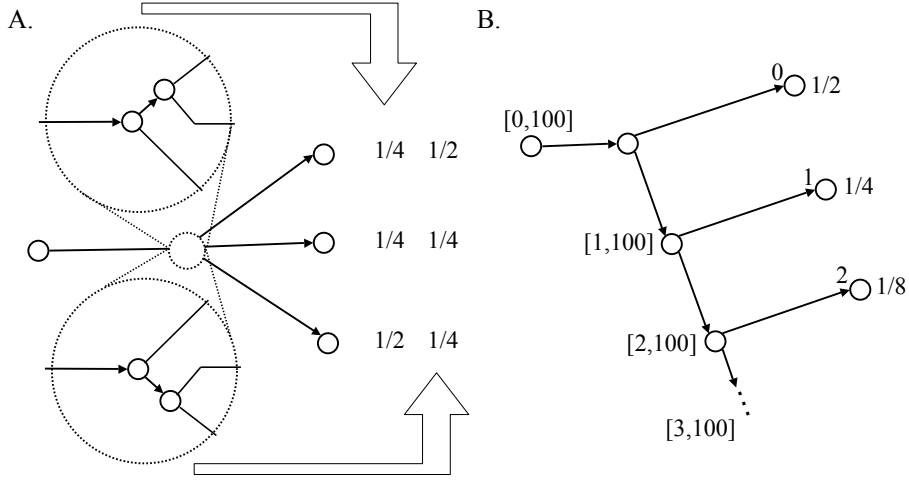


Fig. 3. A. In outcome Egalitarianism, a microscopic change in the details of the Everettian tree has a profound effect on probability. B. The makeup of the Everettian tree results in the probabilistic minimization of the variable. Fractional values represent probabilities.

Fig. 3 illustrates that small changes in the Everettian tree – for example, the order that measurements are made – can have profound effects on probability. It can be readily appreciated that this objection applies to outcome but not branch Egalitarianism. The problem that this creates is that it would appear to give an agent with free will the ability to control probabilities arbitrarily. If an experimenter simply *chooses* to rearrange the order with which measurements are made, then she can wildly affect the outcome of the measurements – and this, so the objection goes, just should not happen. As an example, consider a series of 101 detector elements yielding a particle position P that takes integer values in $[0, 100]$. The measurement is performed by an apparatus and then communicated to the experimenter. First, the apparatus communicates whether the parameter takes value $P = 0$ or $P \in [1, 100]$; if the latter, then the experimenter next determines whether the parameter takes value $P = 1$ or $P \in [2, 100]$; and so on. Outcome counting, applied to the experimenter’s Everettian tree (see Fig. 3B), would tell us that the probabilities that P takes value 0, 1, 2, ... are 50%, 25%, 12.5%, ... i.e. making the lower values the most likely. If the experimenter were simply to “reverse the order” of the P -measurement process (i.e. starting with a $P = 100$ versus $P \in [0, 99]$ measurement), then

the *highest* values would be most likely. In other words, she could “minimize” P or “maximize” P at will, in effect playing God with the laws of physics.

Response to Obj. 3 *The agent does not have arbitrary power to manipulate the details of the Everettian tree.*

A successful Egalitarian model is *expected* to be characterized by features of the Everettian tree that can be manipulated by an agent of free will – but also, the Egalitarian points out, by features that cannot. On the one hand, what we measure or don’t measure – for example, whether or not to make a *welcher weg* measurement as in the double slit experiment – is *expected* in standard quantum mechanics to have an effect on probabilities in a way that defies our classical intuitions. Witness the quantum Zeno effect (W.M.Itano et al., 1990), predicted by standard quantum mechanics and remarkably reminiscent of our P experiment. On the other hand, we should expect the existence of patterns in the Everettian tree that *cannot* be undone purposefully by an agent via free will. In other words, the experimenter’s “free will” to design an experimental protocol as she sees fit does not translate into the ability to dictate *any and all* aspects of the Everettian tree at will. Some details can be changed at the whim of the experimenter; others cannot. So here’s the question: Is there a way to demarcate the one from the other?

It is worth a short aside to illustrate a very interesting technique to investigate issues of this type involving free will and physical law. This technique is to develop a thought experiment in which free will leads to an apparent paradox or some other problem, and then to recreate the thought experiment in a way that retains the paradox but with the element of free will removed. To illustrate this technique, let us take a brief digression from the quantum to the purely classical realm of general relativity (GR). (This digression will turn out to be worthwhile later, when we consider whether the T that we seek for the derivation of the Born rule might be GR.)

6.3.1 *Free will and classical general relativity*

An apparent paradox arises if one considers backward time travel through wormholes containing closed timelike curves (CTC’s) in classical general relativity. Suppose a person traverses a wormhole to go back in time – say 50 years – and kills his grandfather. If he is successful, then he will not have been born, so could not have gone back in time to kill his grandfather ... hence the paradox: the murder happens if and only if the murder does not happen. The only nonparadoxical solution would be for the person to decide not to kill his grandfather, or for some very unlikely event – a strategically placed banana peel, for instance – that results in failure of the murder plot. Either way implies some sort of general relativistic conspiracy. Does this mean that

GR conspires to constrain free will? Or are we forced to assume that GR does not admit CTC's?

A consortium of investigators (Friedman et al., 1990) reasoned that the best way to examine what GR itself has to say about a paradox of this type would be to recreate the paradox, but to remove the element of free will. This was accomplished by Echeverria et al. (1991) in which humans were replaced by a billiard ball (BB). The initial conditions of a typical scenario are such that the billiard ball is traveling along a geodesic in free space towards the entry mouth of a wormhole that is big enough to be traversed by the billiard ball. The ball enters the mouth, traverses the throat and in the process time-travels some small interval *back in time* (in the fixed stars reference frame), and exits the other end of the wormhole along a trajectory such that the ball's entry and exit trajectories cross; that is, the exiting billiard ball is on a *head-on collision course* with the entering billiard ball. (Although this collision would appear to be a single event from the perspective of an outside observer using the fixed stars as reference frame, the billiard ball actually experiences it twice in its frame of reference – say, at proper times τ_1 and τ_2 .) This collision prevents the billiard ball from entering the wormhole in the first place, which means the collision does not take place, which means it does enter the wormhole ... and once again, a paradox, but without complications of free will.

The question is whether there does or does not exist a unique solution to the equations of motion of the billiard ball: in the language of Echeverria et al. (1991), whether the Cauchy problem is “well posed” (one solution) or “ill posed” (no solutions). The grandfather scenario appears on the surface to be ill-posed: no internally consistent solution at all. But in the case of the billiard ball, the surprising discovery was that a solution does exist – in fact, more than one solution! It is as if GR does conspire: every attempt to create initial conditions with “no solution” turned out to have solutions – and not just one, but a *multiplicity* of solutions. Typically, these solutions involved trajectories into and out of the wormhole that were *slightly* different than those originally intended, as well as a self-interaction event different than originally intended; the “head-on collision” was replaced by a “glancing blow,” so to speak. (It is the glancing blow that adjusts the entry and hence exit trajectories ever so slightly.) If we carry the analogy back to the grandfather paradox, then there is the *implication* that *perhaps* the equations of GR do in fact encode a conspiracy one way or another to avoid the grandfather's murder, just as it manages one way or another to avoid the head-on collision. How this might happen is a matter for some very interesting debate that will not be covered here; but see e.g. Everett (2004).

Multiplicity was considered to be a problem: it indicated that the Cauchy problem is ill-posed in an unexpected way: “far too many solutions.” Durand (2002) has argued to reinterpret multiplicity to mean that *GR is a funda-*

mentally non-deterministic theory, and that GR with CTC's can simulate the nonlocal quantum behaviour of simple EPR setups. Later in this paper we will pick up this line of reasoning and argue that multiplicity and Everettian branching are one and the same thing, thereby suggesting that GR with CTC's is the candidate *T* that the Egalitarian seeks.

The original point of this digression has been to argue to be wary of making hasty conclusions about the unworkability of a physical model of nature based on arguments of free will. Such conclusions (e.g., Obj. 3) may be premature without a more careful analysis that just may turn up something very interesting. Who would have thought that an investigation of *free will* (grandfather paradox) might produce a mathematical insight into *indeterminism* of GR?

6.4 The 'Inability to Approximate' objection

Objection 4 *There is no level of approximation at which outcome counting is tractable, because probabilities are exquisitely sensitive to microscopic details of the Everettian tree.*

Considerations of decoherence have taught us that what we model as a simple discrete-valued (eg, spin half) measurement, is in fact a complex process involving a long sequence of interactions between one environmental subsystem and another. So at the utmost level of microscopic detail, the Everettian tree corresponding to even the simplest of experiments is expected to be extraordinarily complex – perhaps infinitely complex. And the proposed answer to issues of continuity and infinity (see Objection 1) was to develop a sampling scheme as a method of approximation. But for any approximate Everettian tree, there will be a higher order approximation yielding a more detailed Everettian tree. The probabilities we assign at one level of approximation will be sensitively dependent on the structure at the next higher level of approximation (see Fig. 3) – and so on, ad infinitum. Thus, says the Critic, no matter how fine-grained we get, there is *no level of approximation at which to apply the outcome counting rule*.

Response to Obj. 4 *Another lesson from classical statistics: There is no reason not to assume that there can exist a threshold level of approximation of the Everettian tree, at and above which outcome Egalitarianism is valid.*

Classical mechanics, as we have seen, is successfully modelled by Egalitarianism. But if the approximation to a classical mechanical system is exceedingly coarse grained, or if it is partitioned using the wrong measure, then Egalitarianism begins to fail. For example: if a system of gas molecules in a container is coarse-grained by equal increments of temperature, then it would be incorrect to assign equal probability to each increment. However, thermodynamics

is successfully modeled by Egalitarianism 1) if we use the correct partitioning, and 2) *once we reach a sufficient level of approximation (degree of fine graining), but not before*. Anything higher than this “threshold” level of fine-graining continues to remain valid. There is no reason not to imagine that *the same is true of quantum phenomena*. The basic idea is illustrated in Fig. 1C, which illustrates the quantum-to-classical transition at threshold 1 and postulates that there exists an attainable level of approximation of sufficient fine graining, threshold 2, at and below which Egalitarianism replaces the Born rule.

A second possibility is that the Everettian tree is not infinitely complex; that there exists some degree of magnification at which all detail of the Everettian tree has been revealed. This is suggested, for instance, by loop quantum gravity that postulates a finite minimum area and volume for spacetime (Nicolai and Peeters, 2006).

6.5 *The epistemic and subjective uncertainty objections*

Wallace (2007, Section 9) puts forth two objections that we should accept “branching indifference” – shorthand for the notion that “the ‘branching microstructure’ of constantly dividing worlds which underlies the macroscopic ascription of weights to coarse-grained outcomes ... is in fact irrelevant.”

Objection 5 *Epistemic objection: Egalitarianism is epistemically impossible to follow.*

Wallace (2007) argues: “to be keeping microscopically detailed track of all manner of branch-inducing events ... is beyond the plausible capabilities of any agent.”

Objection 6 *Subjective uncertainty (SU) objection: From the SU standpoint, the branching microstructure is irrelevant.*

According to the subjective-uncertainty (SU) viewpoint, in a branching event “the agent should expect to experience one or other outcome, but does not (and cannot) know which.” He differentiates this from the objective uncertainty viewpoint, according to which “[b]ranching leads, deterministically, to my having multiple future descendants.”

Response to Obj. 5 and Obj. 6 *The Egalitarian seeks an objective, not subjective, account of probability, and the epistemic objection is irrelevant to an objective account of probability.*

Is it reasonable to require that the laws of physics should not apply to agents

who cannot plausibly (consciously) perform the necessary calculations to implement the law? This argument was made by Bugs Bunny to explain how he remains floating in mid-air after the plank on which he stands is cut out from under him: “I know this defies the law of gravity, but, you see, I never studied law.” Presumably the argument is that Bugs is immune to any physical law if it is epistemically impossible for him to follow (the irony of course being that Bugs is smarter than his adversary). But are we really to believe that different laws apply to agents according to their intelligence? Surely this cannot be taken seriously.

On the other hand, the epistemic objection seems more plausible if the calculations of the “agent” are any type of calculations, not necessarily of the conscious variety (in which case the word “epistemic” is inappropriate). By this line of reasoning, the state-machine ontology of the outcome Egalitarian would suggest that any given observer-state ought to have sufficient computational power to calculate the identity of the state(s) to which it can transition.

We will not go into detail regarding the SU objection, primarily because we agree with Greaves (2004) and Price (2006) that the SU viewpoint (as stated) is in fact incoherent. In any case, Wallace himself admits that an Everettian may reasonably adopt OU over SU. We will note that one of his apparently SU-dependent arguments seems to be a combination of Obj. 3 (the Free Will objection) and Obj. 2 (which assumes branch Egalitarianism). Wallace (2007): “Suppose that someone proposes to increase a million-fold the number of the agent’s descendants who see heads: say, by hiding within the measurement device a randomiser that generates and displays a number from one to 1 million, but whose output the agent does not care about.... Then from the SU viewpoint, this just corresponds to introducing some completely irrelevant extra uncertainty.” This objection has weight if one assumes (as does Wallace) branch Egalitarianism; in this context, a free agent could greatly amplify the probability of a particular experimental result by triggering the randomiser, and she could do this after the fact. But this is not an issue in the context of outcome Egalitarianism, where Wallace’s contention loses its force.

6.6 The ‘decoherence’ and ‘basis problem’ objections

Objection 7 *Decoherence objection*

Greaves (2004) argues that the number of branches is not well defined because: “we are assuming a *decoherence-based* conception of ‘branching’ ... [D]ecomposing the universal state into branches by projections onto a slightly rotated basis can wildly affect the *number* (although not the total weight) of branches enjoying a given Consequence.”

Objection 8 *The basis problem objection*

Greaves (2004) argues: “There is ambiguity over just how *coarse-grained* the division of the multiverse into distinct branches is to be: too fine-grained a division will fail to retain the freedom from inter-branch interference that played a crucial role in selecting the preferred basis in the first place; ...”

Response to Obj. 7 and 8 *The decoherence and basis problem objections assume a model based fundamentally on quantum mechanics, whereas an Egalitarian model seeks to apply the ‘counting something’ rule at the level of a deeper underlying classical model T . At this level, decoherence has a different meaning, and the basis problem does not exist.*

These last two objections miss the point that the ultimate goal of Egalitarianism is to provide an *independent* account of probability. In other words, the ideal Egalitarian scheme should be able to derive quantum probabilities from a fundamental theory that looks nothing at all like quantum mechanics, that does not assume the theory or the mathematics. It should *not* necessarily assume the Schrödinger equation, decoherence, or calculate branches by projection onto a basis in Hilbert space.

6.7 Summary

These objections fail because they are based on the misconception that Egalitarianism has more in common with Deutsch-Wallace decision theory than it actually does: DW assumes the Schrödinger equation and seeks a subjective account of the quantum laws, whereas Egalitarianism seeks an objective account that derives the Schrödinger equation from a deeper classical theory T . This misconception prevents the recognition of parallels to classical mechanics, such as the use of measure theory to solve the problems of continuity and infinity.

7 Can we envision a genuinely independent Egalitarian scheme? The development of a toy model

For the remainder of this paper, we wish to explore the development of an independent Egalitarian account of quantum probabilities. The reason for this exercise is to explore the utility of raising and then responding to the objections in Sec. 6. Some decisions in the construction of this model will be somewhat arbitrary, and so will not be given a thorough justification.

The first step in this process is to make a choice for T . We have already seen

that classical GR with CTC's as presented in Sec. 6.3.1 is nondeterministic, so let us pursue this as T , and adopt the interpretation of multiplicity as giving rise to the Everettian notion of branching: they are one and the same thing. We point out in passing that a scheme of this type assumes a high density of microscopic CTC's to model quantum foam (Hawking, 1978), as discussed in Thorne (1994). Given the fundamentality of GR to this scheme, and the fundamental role of geometry and topology, it is also natural (though not strictly necessary) to adopt the notion that *all is geometry*, according to which there are no matter fields that exist "on top of" a background spacetime geometry; instead, *matter is "built" out of spacetime itself*. In this context we will define an Everettian world $\bar{W}$ as a 4-dimensional manifold with metric g_{ij} . The bar is included to emphasize that it shares the property of the observer $\bar{\psi}$ that it is an entirely classical object. The set of "all possible Everettian worlds" will be designated E .

With this choice of T , we immediately have an advantage over our nineteenth century counterparts. Many of the philosophic difficulties with classical probability stem from the fact that Newtonian mechanics, the underlying T , is deterministic; hence the "paradox of deterministic probabilities." Our model avoids this paradox because GR with CTC's, as we have seen, is nondeterministic. Therefore an Egalitarian account of quantum probabilities from GR would arguably not be on the same philosophic footing as for classical probabilities: it would be on even stronger footing!

These are some lessons we will take from Sec. 6:

- (1) We avoid the epistemic objection, SU objection, decoherence objection, and basis problem objection because our starting point is a classical mechanical theory T . We do not assume the Schrödinger equation, Hilbert space, etc.
- (2) We expect to run into issues of infinity and continuity, but hope that they can be solved as they were solved in thermodynamics (Objection 1). Thus, we should keep the tools of measure theory in mind.
- (3) We must pay special attention to the microscopic details of the Everettian tree (Objection 4).
- (4) We assume observer, outcome, and sub-branch Egalitarianism, assuming we can implement Axioms 1 and 2 which render these three counting methods equivalent. A less appealing alternative is to relax one or both of these axioms, in which case we must justify (ontologically) the rejection of one or more of these counting methods. We do not adopt branch Egalitarianism. (Objection 2)
- (5) We are free to conceptualize thought experiments from the vantage point of any physical object as the observer, and it may be necessary to alternate between 'conscious' observers (like people) and 'not-conscious' observers (like billiard balls, mechanical recording devices, etc.). In this way

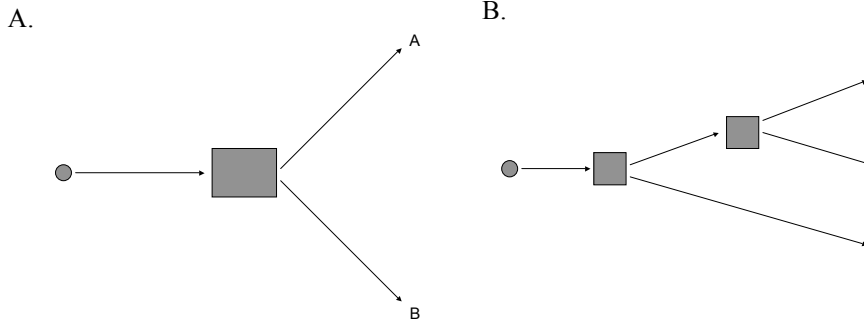


Fig. 4. Multiplicity of solutions in classical mechanics. The circle represents an object such as a billiard ball or a satellite traveling in space along trajectories designated by the arrows. Each box contains a CTC so that entry of the observer into the vicinity of the box results in one of two exit trajectories (multiplicity = 2). A. It is natural to postulate that an observer represented by the circle would expect 50:50 odds of evolution along trajectory A or B. B. A more complicated scenario than that in A.

we may address issues of consciousness when it suits us, but to be able to avoid it when it is necessary to avoid any attendant complications (Objection 3). This flexibility is necessary given the widely divergent opinions on the role of consciousness in quantum mechanics.

7.1 *Branching from multiplicity*

Fig. 4A illustrates an example of multiplicity = 2. The system in question is a billiard ball (the circle) that follows a geodesic through space and interacts with another system (the box) containing closed timelike curves. But let's imagine this time that the billiard ball instead is a satellite, or an astronaut in a space ship, or something else that we are more accustomed to calling an "observer." In a nutshell, our hypothetical toy model takes a diagram of Fig. 4 and interprets it as an Everettian tree. The astronaut becomes the

observer-microstate $\bar{\psi}$; each segment of a trajectory becomes one branch of the Everettian tree. We further assume that each branch is equally “real” and so should receive an equal share of the probability measure. In other words, the astronaut expects a 50% chance to experience movement along trajectory A, and likewise 50% for B. As a slightly more complicated example, consider the more complicated arrangement of Fig. 4B. What frequencies would we expect to be associated with each trajectory? By outcome Egalitarianism, we expect about $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$ for the first, second, and third destinations, respectively.

It should be clarified that multiplicity does not imply that each solution exists together within a single Everettian world $\bar{W}$. If such were the case, then an outside observer would expect to see one satellite entering the box and two coming out. Rather: *no two distinct solutions to the satellite’s trajectory live in the same world*, because if they did, then we would have a violation of the principle of self consistency, defined in Friedman et al. (1990): “the only solutions to the laws of physics that can occur locally in the real Universe are those which are globally self-consistent” (and alternate trajectories are not globally self-consistent). There is no need to assume these manifolds to be “joined” or connected in any way. Example Fig. 4A implies at least two distinct worlds; Fig. 4B implies at least three.

We are led by these considerations to define yet one more counting scheme: world Egalitarianism: all worlds are equally probable. Looking at Fig. 4B, for example, we have three distinct trajectories and therefore (at least) three distinct worlds. Thus we might think that the probability of arrival at any one of the three final destinations is $\frac{1}{3}$. Upon reflection we see that this would essentially resurrect branch Egalitarianism, which we have already rejected! Do we reject world Egalitarianism too, or is there a way to save it? We will argue that we can save it with the following axiom:

Axiom 3 *There is one (and only one) Everettian world for each Everettian sub-branch.*

In Fig. 4B, instead of three worlds, there are four (because there are four sub-branches; compare Fig. 4B to Fig. 2), and we have restored the earlier $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$ predictions. In general, for any observer $\bar{\psi}$, there is an associated ensemble $\Upsilon_{\bar{\psi}}$ of sub-branches $\lambda \in \Upsilon_{\bar{\psi}}$ that flow through it, and by Axiom 3 this defines an associated ensemble of worlds: $M_{\bar{\psi}}$.⁷ With this, we can define:

Definition 5 *World Egalitarianism. Given a branch that splits into N branches, each of which is represented by $\bar{\psi}_j$, each branch inherits a share of probability measure from its parent branch in proportion to the number of worlds in the*

⁷ This is the meaning of the statement: “one can arbitrarily choose a state for one subsystem, and be led to the relative state for the remainder” (Everett, 1957); i.e. where $\bar{\psi}$ is the subsystem, and $M_{\bar{\psi}}$ is the relative state of the rest of the system.

ensemble $M_{\bar{\psi}_j}$.

World Egalitarianism, like outcome Egalitarianism, is transitive.

Axiom 3 allows us to conclude that world Egalitarianism is equivalent to sub-branch Egalitarianism:

$$\begin{array}{c} \text{sub-branch Egalitarianism} \\ \Updownarrow \text{Axiom 3} \\ \text{world Egalitarianism} \end{array} \quad (3)$$

– and hence to outcome and observer Egalitarianism as well.

One might object to Axiom 3 on the grounds that there is really no justification for N *identical* sub-branches (for example, the two branches in Figure 2B leading to the third final destination) to imply N worlds. Why should we require *two* worlds to represent just *one* trajectory? Our answer will be to postulate:

Axiom 4 *No two sub-branches are globally identical.*

This does not mean that two sub-branches cannot be identical *along a segment*, such as the two leading to the third destination in Fig. 4B. We are referring in Axiom 4 to sub-branches λ *in toto*, i.e. *along the entire length*. What do we mean by ‘the entire length’? When we first conceptualized a sub-branch, we reasoned the Everettian sub-tree to be determined from the Everettian tree. To construct the sub-branches for $\bar{\psi}$ in its entirety, we must *complete* the Everettian tree – e.g. continue the trajectories leading to and away from $\bar{\psi}$ infinitely far into its future and its past. We assume each sub-branch either to be infinite in length, or to form a closed loop (meaning it returns to its original state).

7.2 A closer look at the Everettian tree

We want to provide as many details as we can about the nature of the Everettian tree – especially at its most *microscopic* level, for it is *here* that the Egalitarian model meets its most difficult challenges (see e.g. Objections 1 and 4). In Figure 4A we have approximated multiplicity = 2. But are there any *microscopic details* that might throw off our calculations, along the lines of Objection 4 and Fig. 3? Are we able to say definitively what the Everettian tree looks like *exactly*? If not, then can we devise some approximation scheme?

We first realize that for the billiard ball, we cannot define the Everettian tree

exactly. The branch event occurs at the instant of self-interaction τ_1 , which is the moment that the billiard ball “observes” the outcome A or B; put another way, it *experiences* A or B. But consider the fact that *the self-interaction is not felt simultaneously by every particle throughout the ball*. The self-interaction event comes in the form of a collision, and the particles nearest the impact will “feel” the collision *before* the particles on the opposite end of the ball. Therefore, we cannot pinpoint when, *exactly*, the branching process occurs; it must occur over some finite time interval.⁸ Furthermore, this problem will generalize to *any extended object*, no matter how small! So what do we do? Do we need a separate tree for each particle?

With some consideration we realize that, in a way, the answer is: yes. We originally set out to increase the grain of the tree for an observer $\bar{\psi}$, but it turns out we need to do something subtly different: rather than making a more accurate tree for a given observer, we pick a *smaller observer*. More precisely, if we start with $\bar{\psi}$, then we increase the grain of the Everettian tree by changing focus to a *piece* of $\bar{\psi}$. Following this line of thinking to its logical conclusion, we find ourselves forced to ask: *what is the most microscopic piece of matter we can think of?* And this is why we picked the “all is geometry” view: it allows a mathematically precise answer to this question! The smallest and most elementary “chunk” of matter is an *arbitrarily small*, simply connected 4-ball, with the metric g_{ij} fully specified throughout the 4-ball; we designate it $\bar{\psi}^e$ and the collection of “all possible $\bar{\psi}^e$ ” is labelled Ω^e . We remark that any (classical) physical object – particle, billiard ball, particle detector, human brain – can be constructed using elements of Ω^e as building blocks. The superscript e is meant to indicate it is the smallest and most *elementary* observer. We will return to consider the issue of macroscopic (and multiply connected) objects, including Tegmark’s SAS’s, as observers at a later time.

The space Ω^e can be parametrized in the following manner. Specification of the metric g_{ij} throughout $\bar{\psi}^e$ allows the calculation of the metric and power series at any interior point p . Conversely, provided with the metric and its complete power series *at the single point p and nowhere else*, the method of analytic continuation allows the calculation of g_{ij} throughout the entire 4-ball. Thus, we are able to parameterize Ω^e as an *infinite dimensional state space*, with each dimension being one component of the metric or its power series, and $\bar{\psi}^e$ being one point in this space. It is interesting to revisit Wallace’s “epistemic” objection (Obj. 5) with this definition of the observer in mind. To specify $\bar{\psi}^e$, it is necessary to specify an infinite number of parameters, each of which is

⁸ Further difficulty arises when we consider that the timing of the branch process presupposes a frame of reference. But due to thermal movements, different particles exist in different reference frames, so whose do we use? From a relativistic standpoint, simultaneity is not even uniquely *defined* – not when we need to define it *exactly*!

(presumably) a real number. Thus the informational capacity of $\bar{\psi}^e$ is quite large, conceivably large enough to imagine sufficient computational power to calculate the identity of the state(s) to which it can transition.

We remark that Ω^e is simply connected. It is also a *continuous* space, thus raising the issue of Objection 1. The definition of a measure over Ω^e is outside the scope of the present paper; however there is no reason that one cannot be defined, analogous to the measure over the space of microstates in classical statistical mechanics discussed earlier.

7.3 Time-dependent state evolution

In classical general relativity, the evolution in time of a physical object is described by its propagation along a geodesic, as calculated by the parallel transport law, i.e. according to the geodesic equations (Misner et al., 1973):

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt} = 0 \quad (4)$$

These, of course, are the equations that govern the trajectory of the billiard ball in Friedman et al. (1990). They govern the evolution of any piece of matter, including the elementary observer $\bar{\psi}^e(\tau)$ within any given spacetime, where τ is the proper time of the observer.⁹

Within each spacetime $\bar{W}$, there will be a distinct evolution $\bar{\psi}^e(\tau)$ calculated using the geodesic equations; this is a geodesic through $\bar{W}$, but also yields a curve through state space Ω^e .¹⁰ (The set of all possible geodesics defines the set of all possible sub-branches λ .) For any particular $\bar{\psi}^e(0)$, there is a uniquely determined set of all possible evolutions. If these evolutions are drawn on a graph with location in Ω^e as the ordinate and τ as the abscissa, the result is the Everettian tree.¹¹ For any point $\bar{\psi}^e(0)$ in Ω^e , there is a *unique, well defined* Everettian tree rooted at that point. The Everettian tree therefore “lives” in Ω^e .

⁹ It should be remarked that a complete definition of the elementary observer $\bar{\psi}^e$ includes a unit vector pointing in (defining) the forward time direction in addition to specification of the metric and power series.

¹⁰ Typically, any two evolutions $\bar{\psi}_i^e(\tau)$ and $\bar{\psi}_j^e(\tau)$ may be identical up to some time τ_b , where b indicates “branch point,” but nonidentical after.

¹¹ A less complete construction might use only the ten components of g_{ij} instead of the infinite dimensions of Ω^e as the ordinate.

7.4 Probabilities for ‘self-aware states’

We have outlined an exact (in principle) method to draw an *infinitely fine-grained* Everettian tree using elementary observers $\bar{\psi}^e$ as states. With the Everettian tree (and sub-tree) drawn, we calculate probabilities using any one of four equivalent counting methods:

$$\begin{array}{rcl}
& \text{observer Egalitarianism} & \\
& \Downarrow \text{Axiom 1} & \\
& \text{outcome Egalitarianism} & \\
& \Downarrow \text{Axiom 2} & (5) \\
& \text{sub-branch Egalitarianism} & \\
& \Downarrow \text{Axiom 3} & \\
& \text{world Egalitarianism} &
\end{array}$$

where $\Downarrow$ indicates equivalence of counting methods.

But we still have not specified the principle from which to calculate probabilities for macroscopic observers, in particular, macroscopic observers that are SAS’s: $\bar{\psi}^{SAS}$. The difficulty is that the Everettian tree for a macroscopic observer has not been exactly defined. Here are two techniques:

7.4.1 Approximation method

The first method, the *approximation method*, is to use an embedded microscopic observer as a proxy for the macroscopic object of interest. We can then use any of the counting techniques discussed so far to calculate probabilities.

7.4.2 Exact method

The second method is to use observer Egalitarianism. Of all counting methods, this is the only one that does not require us to construct the Everettian tree. All that we must construct is U , which we defined earlier as the set of all possible observers, with the number of copies of a given observer being proportional to its measure. We have seen how to do this with elementary observers. Our next task is to calculate the measure of an observer that is not elementary, and to do this we must construct U for macroscopic observers. Recall the earlier comment that any object can be constructed using elementary observers as building blocks. Let us augment Axiom 1 with the following:

Axiom 5 U is a unique, infinitely nice (smooth, continuous, analytic met-

ric) manifold. Every possible physical object, of any size, is found somewhere embedded in U .

We know that elementary observers are (or can be, in principle) represented more than once in U , and there is no reason that it will be different for macroscopic observers. Given any piece of matter we can define its measure as the number of times it occurs in U . In this way, we have an exact definition for probabilities for macroscopic observers, even though we do not have an exact Everettian tree for a macroscopic observer!

We may assume that Axiom 5 applies to 4-dimensional Everettian worlds $\bar{W}$, which we may imagine to be the most macroscopic physical objects that exist. In other words, from Axiom 5 we may deduce that the set of all possible worlds $\bar{W}$ is the set of all possible 4-dimensional hyperslices through U .

8 Simulation of quantum mechanics

Durand (2002) argued that GR with CTC's can simulate quantum nonlocality. Although his arguments were not quantitative, he was able to construct a model of EPR-like entanglement between two particles. In his simulation, the entangled particles were connected by CTC's. In this section we will construct a more elaborate thought experiment to show that this model can simulate nonlocality. Two features are noteworthy. First, it is not necessary that the observer do the time-traveling; unlike Fig. 4, in this section it is the observed system, not the observer, that traverses the wormhole. Second, we will not require a wormhole to connect the distant particles to simulate entanglement. Our goal will be to design a series of increasingly complicated thought experiments until we simulate the double delayed choice (DDC) Einstein-Podolsky-Rosen (EPR) gedanken experiment measuring the spins of a pair of entangled electrons.

Floating in space, unmoving with respect to the fixed stars frame of reference, there exists an apparatus composed of a small wormhole, a small billiard ball (BB), and a device to throw the BB into the wormhole, detect which of N trajectories is the result, and retrieve the BB for another throw. Any desired value of N (finite, integral) can be achieved by varying the initial trajectory. There is also a computer that controls the device, stores the results of each iteration, and communicates the results of such experiments to distant observers using flashes of light. To signal a binary result, a flash that is either red or blue is emitted. To signal more information, a single flash that is one of N possible frequencies may be emitted. Alternatively, a series of binary flashes may be emitted. Any given flash of light may be emitted in all directions, or alternatively may be directed along a desired axis through the use of a barrier

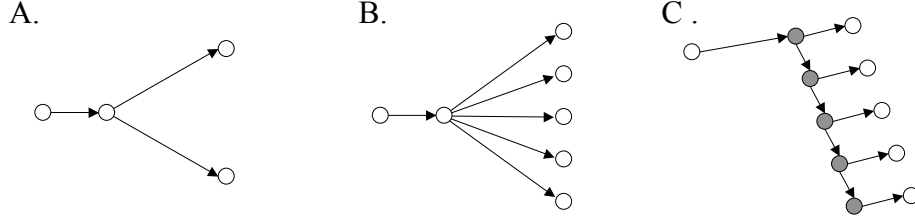


Fig. 5. Everettian trees corresponding to various signaling methods. See text for details. In C, each shaded state corresponds to receipt of one binary light pulse.

to block the light from going in unwanted directions.

Alice and Bob are situated a very large distance away from the apparatus at $-D$ or $+D$ respectively along the z -axis. If the computer performs a single iteration of multiplicity 2 (like a coin flip; see Fig. 6A) and immediately signals the result to Alice and Bob with a single flash of red or blue light, then Alice's pre-observation state will evolve into the observed-red or observed-blue state at the instant the light pulse reaches her. This corresponds to the split in her Everettian tree (Fig. 5A). Bob's will look identical, and when Bob and Alice later confer they must always agree on the results of any given experiment (this is the principle of self consistency). This is a simulation of a simple EPR experiment in which a pair of electrons is emitted in opposite directions, with parallel spin measurements performed on each electron. We will next consider a variety of signaling methods of increasing complexity that the computer can use.

- The apparatus signals Alice but not Bob, and Alice immediately retransmits the result in the form of a flash directed towards Bob. In this case then Bob's Everettian tree will look just like Alice's, except that the split does not occur until Alice's signal reaches him. (*Retransmission* of a result from Alice to Bob or vice versa is equivalent to the two of them getting together later to

confer about their results.)

- The previous method can also be performed using a value of $N > 2$, with the results signaled by a single flash of light in one of N frequencies. In this case, Alice and Bob's respective Everettian trees take the form of Fig. 5B (for $N = 5$), so that (by outcome counting) each result is given probability measure $1/N$.
- The computer signals the result n (to both Alice and Bob) using a single red vs. blue light bulb that emits *a series of N binary pulses*; each pulse is red except for the n^{th} pulse, which is blue. The corresponding Everettian tree is shown in Fig. 5C (for $N = 5$). (See also Fig. 6B, for $N = 6$.) By outcome counting, the lowest value of n is much more probable; n has been minimized. (As we discussed earlier, this provides a simulation of the quantum Zeno effect.)
- The apparatus contains a series of N binary flash bulbs that are arranged in order from 1 to N on a stick that lies parallel to the z axis, with results directed only in Alice's direction. The computer triggers each of the N bulbs to flash *simultaneously* (in the fixed stars reference frame), with all flashing red except for the n^{th} which flashes blue. Light bulbs corresponding to lower values of n are closer to Alice, and her Everettian tree is once again Fig. 5C, so that n is minimized.
- The apparatus contains a signaling device composed of N light bulbs equidistributed on a circular ring from 0 to 360 degrees, with the plane of the circle being perpendicular to the z -axis. Once again the computer triggers each of the N bulbs to flash *simultaneously* (in the fixed stars reference frame), with all flashing red except for the n^{th} which flashes blue, and the flashes directed only towards Alice. Since Alice is also on the z -axis (at $-D$), then each flash will reach her simultaneously, so that the corresponding Everettian tree is Fig. 5B. However, if Alice moves any distance away from the z -axis at an angle θ in the xy plane, then the flashes of light will reach her sequentially rather than simultaneously, with the flash from the light bulb closest to angle θ reaching her first. She can predict that she will observe the blue light to be on the circle at (or very close to) the angle θ . If there are 360 light bulbs, then she should expect $n = \theta$ (or very close to it). Her method of measurement affects the measurement itself – which, as we have discussed earlier, is a necessary feature of any model of quantum mechanics. Incidentally, this feature illustrates the inherent nonlocality of the toy model: by changing the method of measurement over “here,” Alice instantaneously changes the most likely value of a variable over “there,” e.g. the pattern of the light bulb flashes observed on the opposite side of the apparatus at $+D$ (in the case that the flashes are emitted in all directions).

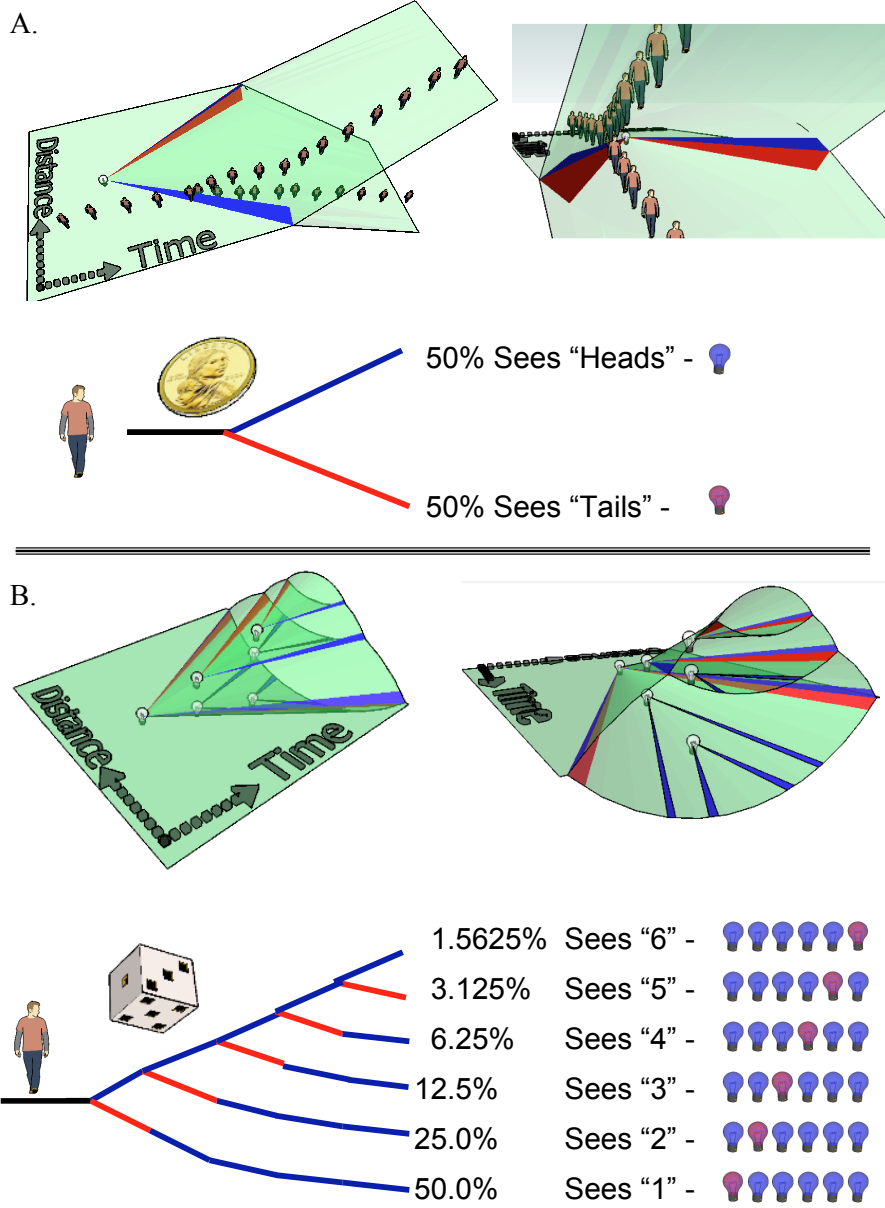


Fig. 6. A. A coin is flipped. The result (heads or tails) is signaled to a distant observer using a single flash of light (red or blue). The expanding light cone defines the separation of one Everettian world from the other. The Everettian tree is shown. By outcome counting, each result is equiprobable. B. A six-sided die is rolled. The result $n \in [1, 6]$ is signaled to a distant observer using six binary flashes of light: all are blue except for the n^{th} , which is red. (The first three flashes are shown). The Everettian tree is shown. By outcome counting, 1 is more likely than 6; it has been minimized. Reversal of the order of flashes would result in its maximization.

8.1 Simulation of the double delayed choice EPR gedanken experiment

The double delayed choice EPR gedanken experiment involves the emission of an entangled pair of unpolarized electrons from the origin to Alice and Bob who are situated far away on the z axis at $-D$ or $+D$, each of whom operates a Stern-Gerlach apparatus whose measurement setting angles θ_A and θ_B they are free to set anywhere between 0 and 360 degrees (hence the “choice;” “double” because they both have choice). This angle can be set just prior to receipt of the electron, thereby preventing Alice or Bob from being aware of the other’s angle setting at the time of the measurement via classical signal propagation due to speed of light restriction (hence the “delayed”). On any single iteration, Alice may expect a 50:50 chance to see spin up, and will expect Bob to observe the opposite spin with probability $\cos^2(\frac{\theta_A - \theta_B}{2})$. Bob expects likewise.

For this simulation, the computer conducts five independent iterations with multiplicities 360, 360, 100, 100, and 2, generating five integers $N_1, \dots, N_5$ (a total of 2,592,000,000 distinct combinations). These are recorded in the computer which then analyzes the data and emits the following series of signals (sequentially, but in a tight packet). First, the values of N_1 and N_2 are each independently signaled by a circular ring of 360 light bulbs (Fig. 5C) to Alice and Bob, respectively. Next, N_3 and N_4 are each independently signaled using a single flash (one for each) that takes one of a possible 100 frequencies (Fig. 5B) to Alice and Bob, respectively. Next, if $N_5 = 1$ (or 2) then a single red (or blue) pulse is signaled to Alice. Simultaneously, the *opposite* signal (blue or red) is sent to Bob if, by the computer’s calculations, $\frac{|N_4 - N_3|}{100} \leq \cos^2(\frac{N_2 - N_1}{2})$ (interpreting N_1 and N_2 as being in units of degrees); otherwise, Bob receives the *same* colored signal.

What do Alice and Bob expect to observe? Let us analyze the situation from Alice’s perspective. Immediately prior to receipt of her packet of light pulses (N_1 , then N_3 , and finally a red or blue pulse to represent spin up or down), she positions herself a small distance away from the z -axis, at arbitrarily chosen angle θ_A in the xy plane. As discussed above, she expects the following: 1) that the signal for N_1 (or N_2 , for Bob) approximately equals (in degrees) her chosen angle θ_A (or θ_B , for Bob); 2) that the signal for N_3 (or N_4 , for Bob) corresponds to an integer in $[1, 100]$, each with probability $1/100$; and 3) that her and Bob’s spins are opposite iff $\frac{|N_4 - N_3|}{100} \in [0, 1]$ is less than or equal to $\cos^2(\frac{N_1 - N_2}{2})$; this happens with probability $\cos^2(\frac{\theta_A - \theta_B}{2})$. They test these predictions in the usual manner: by running an arbitrarily large number of spin measurements, after which they sit down together and analyze all available data.

In every respect, qualitative and quantitative, this toy model thought experiment is a perfect simulation of the quantum mechanical version. When they compare their data, they will each confirm their predictions: that for each it-

eration, N_1 and N_2 take values equal (approximately) to the chosen settings θ_A and θ_B ; that the values for N_3 and N_4 appear equidistributed in the interval $[0, 100]$; that each person observes spin up and down approximately 50 percent of the time, with no way to predict any individual spin value prior to measurement; and that their spin up and spin down values are correlated using the familiar cosine-squared law. Importantly, at the completion of any single iteration there is no way for Alice (or Bob) to predict ahead of time (prior to their meeting) the value of N_2 (or N_1), the other person's angle θ_B (or θ_A), or the other person's spin result. Of course, the simulation supposes the existence of "hidden" variables N_1 , N_2 , N_3 and N_4 that have no counterpart in standard QM. The simulation is therefore a hidden variable model, one that is not excluded by Bell's theorem because of its inherent nonlocality (and Bell's theorem excludes local but not nonlocal hidden variable theories).

The ability for the toy model to simulate the DDC EPR experiment is not intended as a derivation of quantum statistics, but is offered as indication that a derivation may be possible. It is noteworthy that an independent proposal to model the spin-half properties of the electron has been proposed by Hadley (2000) that, like the spin simulation just described, is based on a model that relies fundamentally on CTC's to derive quantum mechanics from GR (Hadley, 1997).

8.2 *Critiques of the simulation*

There are several critiques of the above model that must be considered. First, we are assuming that the results of a given iteration can only be communicated to Alice and Bob via the light pulses, and that there is no other type of interaction between the apparatus and the environment that perturbs the state of Alice and thereby communicates the result. However, we would expect that gravitational waves could work in the same way as a light pulse: each of the two possible BB trajectories will emit a unique gravitational wave "signature." Once these waves reach Alice, the pre-observation Alice must evolve into the "Alice with BB moving this way waves" or "Alice with BB moving that way waves" state. This is so despite the fact that these gravitational waves will be extraordinarily miniscule in amplitude. On the other hand, considering that our model is based on the assumption of highly nontrivial spacetime topologies, we could imagine that either no waves are emitted (say, shielded by some sort of CTC-associated event horizon) or that the gravitational waves emitted are exactly identical for each of the N trajectories and therefore do not induce the splitting of Alice into multiple states. In other words, we can postulate two types of information associated with events of multiplicity > 1 : type 1 which leaks immediately into the environment in the form of gravitational waves and thereby induces splitting in any and every observer in its future light cone,

and type 2 which resides in some more restricted volume of spacetime, does not leak into the environment, and does not induce such rampant splitting. It is beyond the scope of this paper to provide a mathematical description of the difference between type 1 and type 2 information in any sort of detail. In the toy model we have assumed the computer to have the ability to assign information arbitrarily into the type 1 or type 2 category (by choosing whether any given light pulse is emitted in every direction for all to see, or sent in a directed fashion to one particular observer), and in this way, the simulation is contrived.

The second critique is that some kinds of type 1 information in the model can lead to unphysical situations. For example, suppose that the simultaneous N binary flashes emitted from the stick parallel to the z -axis are emitted in all directions instead of only to Alice. In this case, the model predicts that Alice should expect to witness the minimum value for n , while Bob should expect to witness the maximum value for n . When they get back together, though, the principle of self-consistency requires that within any individual 4-manifold $\bar{W}$, they must agree on what they saw. Therefore, Alice predicts not only to witness the minimum value of n , but also that she will later meet with a stunned Bob who is surprised that n has taken what he expected to be its most unlikely value. This *expectation of surprise* from a different observer is what makes it an unphysical, *pathologic* scenario; it is found neither in classical mechanics, nor in quantum mechanics. In other words, pathology is inconsistent with both classical and quantum mechanical laws.

A simple way out of this difficulty is to allow pathologic scenarios to exist in principle, but to require the exclusion of pathology – perhaps a in low energy limit – via the relegation of certain types of information to type 2 status. In other words, we require the exclusion of pathologic scenarios to be a necessary requirement of the low energy limit, but not necessarily excluded under conditions of sufficiently high energy that quantum mechanical laws break down. How to do this is beyond the scope of the present paper. However, it will be speculated that a solution may involve the identification of an *action principle*, in which a particular quantity is minimized (or maximized) via methods described in this paper. It is well known that many important results in physics can be derived via an action principle, well established examples including Newton’s laws (Marion and Thornton, 1988) (via Hamilton’s principle of least action) and the Einstein equations (Misner et al., 1973). It should be noted that action principles are typically introduced as a *freestanding axiom*. The demonstration of an action principle as a *consequence* of a probability rule (outcome counting) would be rather unique, and could offer a powerful method for the definition of the ‘low energy limit,’ the elimination of pathology, and (hopefully) the derivation of the Schrödinger equation and the Born rule.

9 Discussion

The central claim of this paper is that the Egalitarian account of quantum probabilities remains a viable option as a solution to the problem of probability. Ultimately, the question of Egalitarianism’s coherency can only be settled in one of two ways: (1) that it is incoherent, via a rigorous no go theorem, which we argue has *not* been provided; or (2) that it is coherent, via a proof-by-existence, i.e. a specific Egalitarian theory. In the absence of (1), the dismissal of Egalitarianism by advocates of decision theory is profoundly premature.

If some day string theory should succeed as a successful theory of Nature, and if it should retain the Egalitarian concept of probability as discussed in Sec. 4, then (2) will have been accomplished. It is interesting to speculate that the “philosophic” considerations surrounding Egalitarianism might therefore provide some insight into the interpretation of string theory, which is sometimes criticised on the grounds that its physical interpretation does not yet match its mathematical sophistication – i.e., that it is not yet a theory at all.

The use of philosophic or ontologic considerations as a guide for model building in a fashion beyond the mere interpretation of an existing theory is a potentially fruitful, yet difficult, endeavor. Regardless of whether the toy model proposed in Sec. 7 is right or wrong, it is hoped that it may serve as an example to demonstrate how philosophic considerations may inform model-building. Indeed, this toy model is motivated specifically to address the problem of probability; the persistent unease with the Deutsch-Wallace decision theoretic as well as previous attempts to formulate a self-interpretation of the MWI ought to tell us nothing if not that such a model should be sought.

The proposed toy model begins with the assumption of GR with CTC’s as the fundamental theory T , and aims to derive not only the Schrödinger equation – which, recall, is *assumed*, rather than derived, in Deutsch-Wallace decision theory – but also the Born rule. Although this is not achieved fully in this paper, it is demonstrated (albeit with some contrivances) that quantum statistics can be simulated *quantitatively* in the case of the EPR experiment, considered to be one of the definitive demonstrations of quantum nonlocality. It is speculated that there may be a way to define an action and implement an action principle to demonstrate (without contrivances) quantum statistics in (perhaps) the low energy limit; again, this is well beyond the scope of the present paper. Many of the choices in the construction of this model are arbitrary, including the first step, the choice for T . There are at least two additional attempts in the literature to build an Egalitarian model, independent from this one, each of which proposes a scheme to construct an Everettian tree along the lines of Fig. 1B. In the first model, Hanson (2003) proposes to envision

a pruning process that begins with a large number of branches and eliminates them until a number proportional to $|\phi_n|^2$ remain. In the second model, Weissman (1999) proposes the reverse: to envision a process of splitting that eventually generates (over some small time interval) a number of branches proportional to $|\phi_n|^2$. Although these appear to be based on the problematic branch-Egalitarianism, they could easily be reconceptualized in terms of the more favorable sub-branch Egalitarianism with no alteration in their essential structure.

It is likely that the toy model will be criticized on the grounds that it is too ambitious, because it attempts to deal with too many difficult issues all at once. These issues include:

- (1) The problem of probability.
- (2) The paradox of determinism.
- (3) The “all is geometry” notion. The idea that matter is built out of space-time is as old as GR itself. This idea, however, has unfortunately been assumed (prematurely!) to fail by virtue of the fact that no one has made it succeed, despite eight decades of valiant attempts: Einstein-Rosen bridge (Einstein and Rosen, 1935), Wheeler’s geons (Wheeler, 1962), etc.
- (4) Do CTC’s actually exist? Some might argue that if we take GR seriously, then the answer must be yes; this is the stance adopted for this paper. On the other hand, their existence raises problematic issues regarding the nature of causality, free will, etc. Furthermore, there is no experimental evidence for wormholes. Their existence remains controversial; many favor Hawking’s Chronology Protection Conjecture which simply states by *fiat* that Nature does not allow them (Thorne, 1994).
- (5) The role of the observer.

We respond by asking the Reader to consider the possibility that these issues cannot be solved one at a time. Perhaps, as the toy model postulates, each issue is intimately connected to the others so much so that they can *only* be solved simultaneously. Indeed, our starting position was that issue number (1) could not be solved without addressing issue (5) simultaneously; the failure to do so runs us afoul of Bell’s Theorem. We saw also in Sec. 6.3.1 that the attempt to solve a paradox involving free will (yet another difficult issue) resulted unexpectedly in an argument that GR is nondeterministic, thus addressing issue (2).

What is the justification for Egalitarianism in the first place? The complexity of this question is underscored by the identification in this paper of at least four different counting schemes, each one of which will likely have its own appeal among different Readers. For the purpose of this paper, we will simply remark that by relationship (5), which states their mathematical equivalence, the acceptance of one on ontologic or philosophic grounds implies that they

are all acceptable.

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